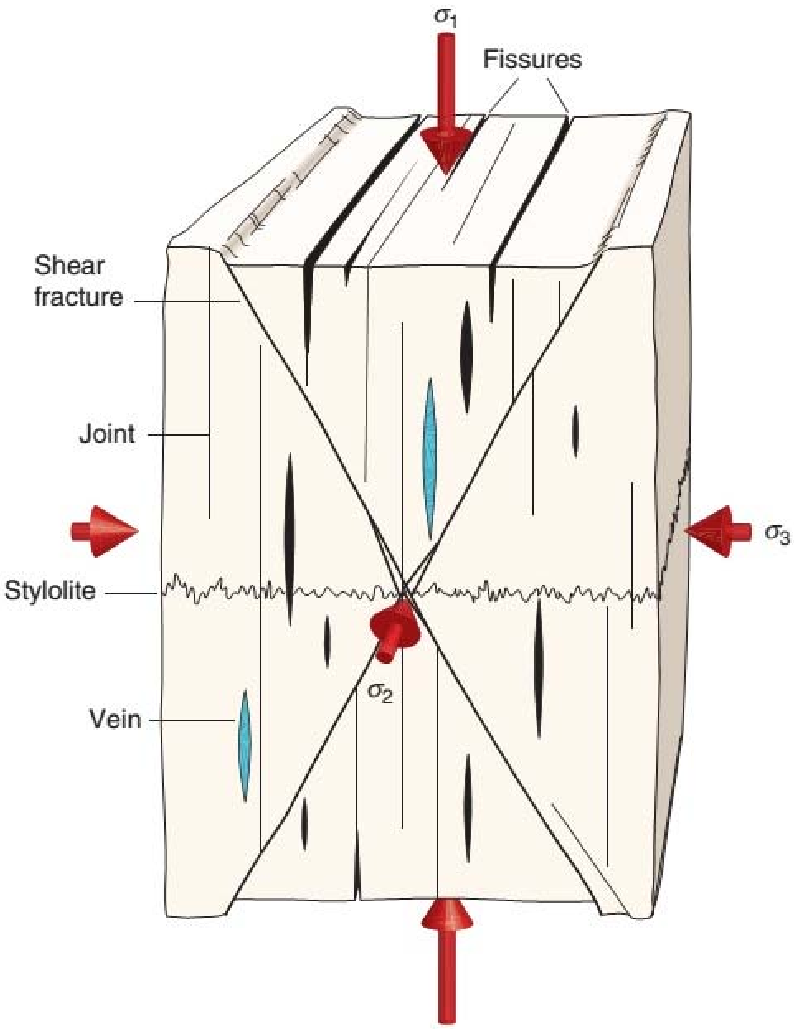
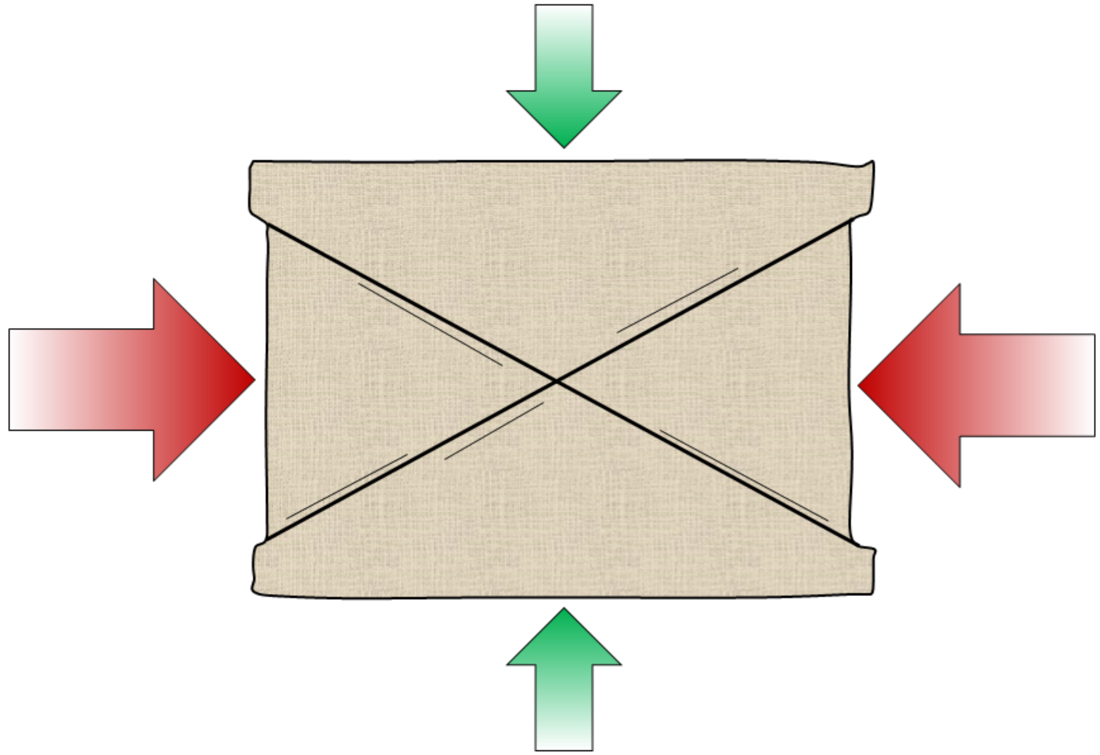
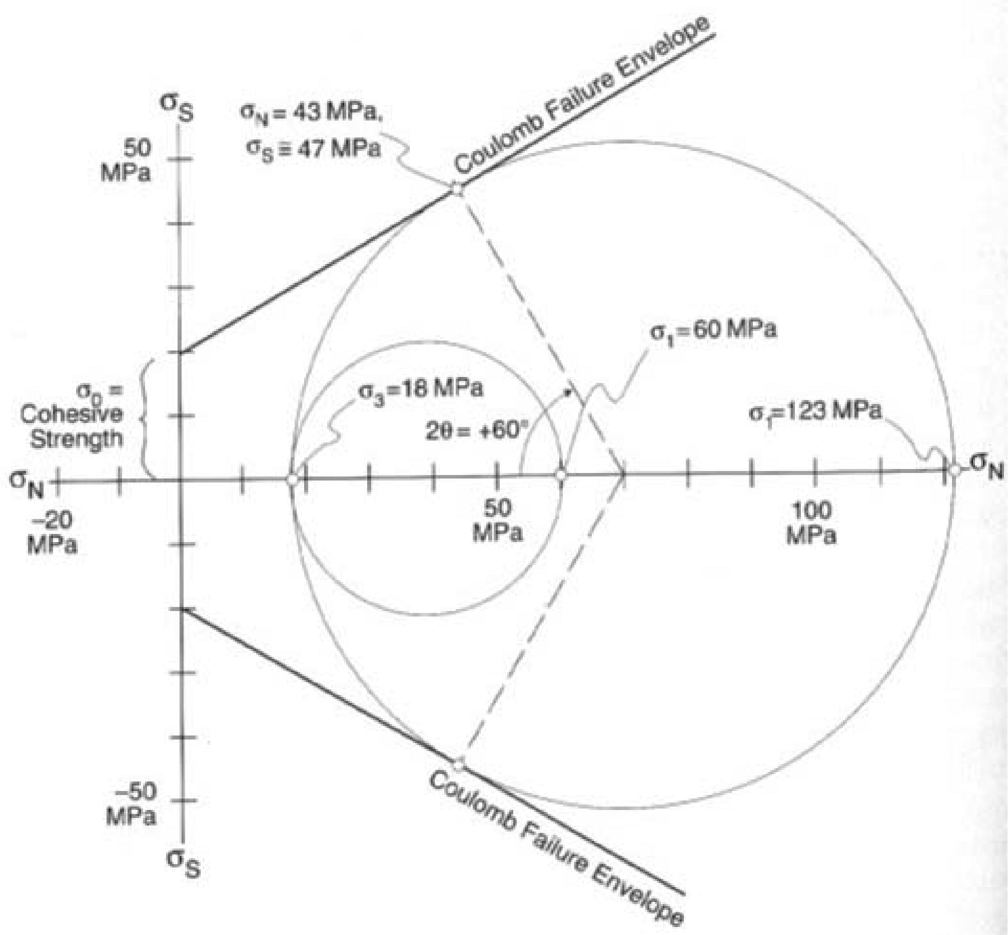
FAULTS



σ_1
Fissures
Shear fracture
Joint
σ_3
Stylolite
σ_2
Vein





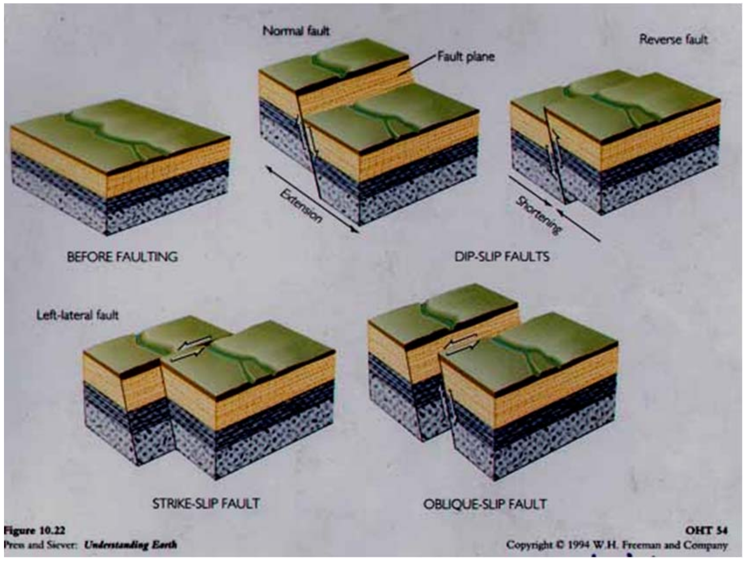
- The cohesive strength (σ_0) is a small part of critical shear stress required for shear fracture.
- Most shear fractures form when shear stresses on a plane of failure reach a level slightly over 50% of the normal shear stresses acting on the plane.

FAULT CLASSIFICATION AND TERMINALOGY

Faults: ***Are fractures that have appreciable movement parallel to their plane. They produced usually be seismic activity.***

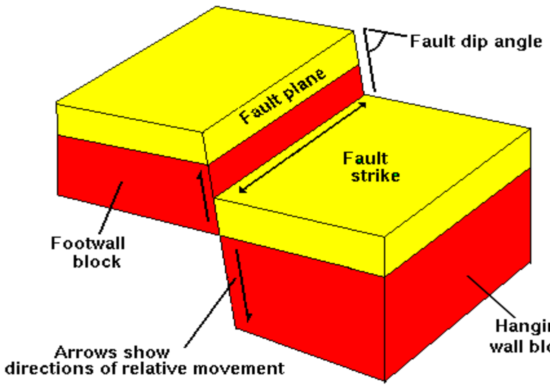
Understanding faults is useful in design for long-term stability of dams, bridges, buildings and power plants. The study of fault helps understand mountain building.

Faults may be hundred of meters or a few centimeters in length. Their outcrop may have as knife-sharp edges or fault shear zone. Fault shear zones may consist of a serious of interleaving anastomosing brittle faults and crushed rock or of ductile shear zones composed of mylonitic rocks.

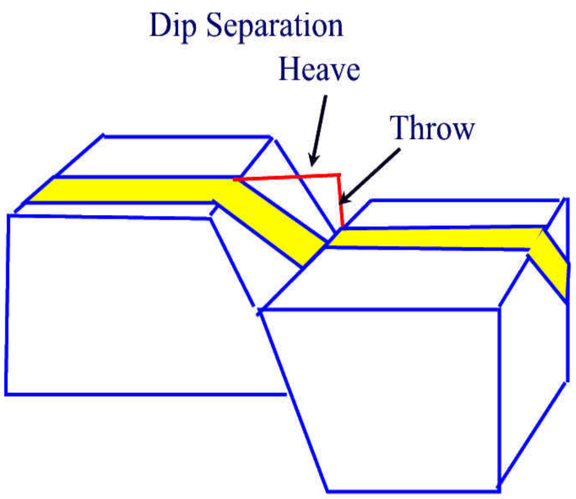


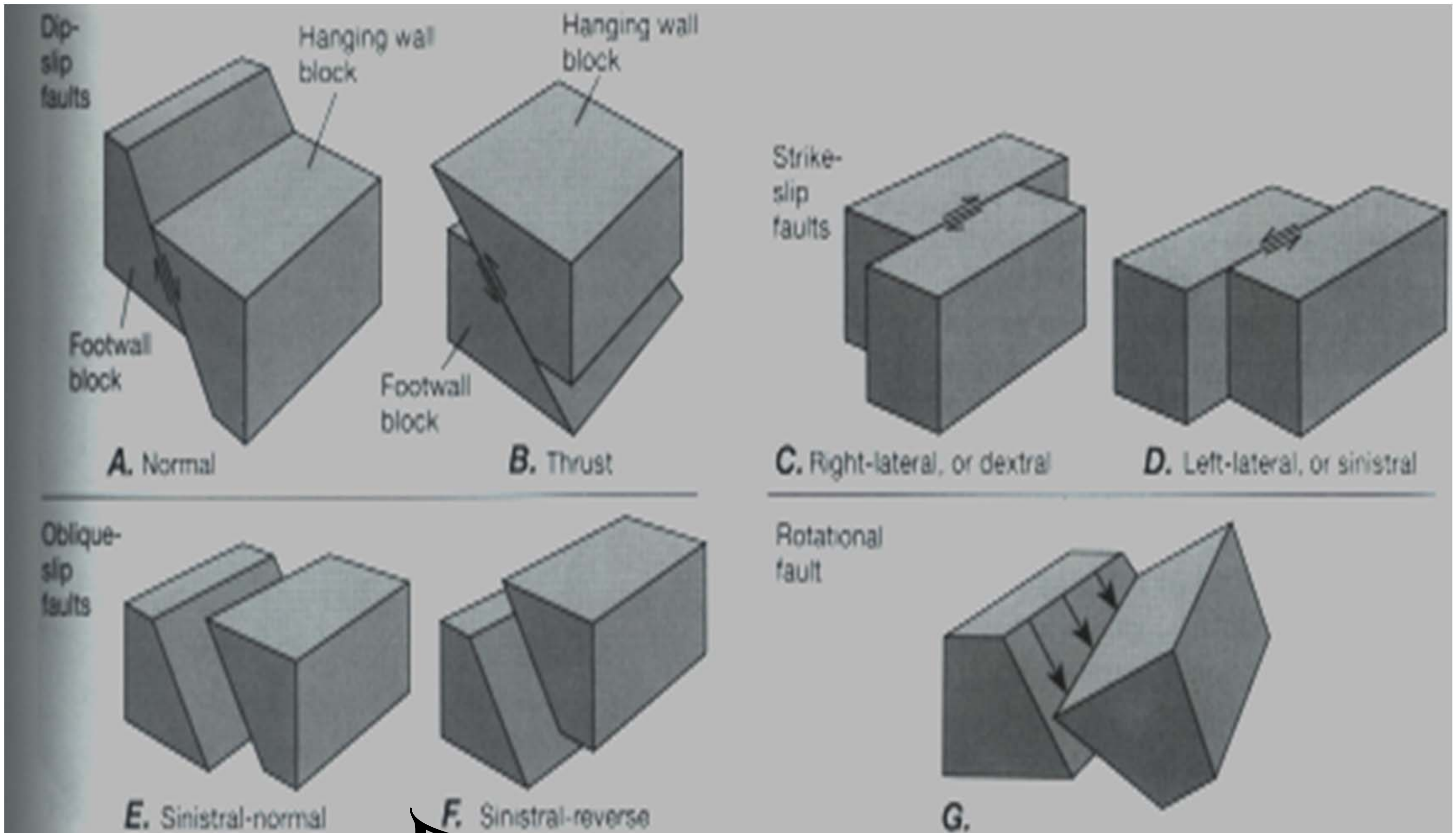
Parts of the Fault

- **Fault plane**: ***Surface that the movement has taken place within the fault.On this surface the dip and strike of the fault is measured.***
- **Hanging wall:** ***The rock mass resting on the fault plane.***
- ***Footwall:*** ***The rock mass beneath the fault plane.***
- **Slip:** ***Describes the movement parallel to the fault plane.***
- ***Dip slip: Describes the up and down movement parallel to the dip direction of the fault.***
- **Strike slip:** ***Applies where movement is parallel to strike of the fault plane.***
- **Oblique slip:** ***Is a combination of strike slip and dip slip.***
- ***Net slip (true displacement): Is the total amount of motion measured parallel to the direction of motion***



- **Separation:** ***The amount op apparent offset of a faulted surface, measured in specified direction. There are strike separation, dip separation, and net separation.***
- **Heave:** ***The horizontal component of dip separation measured perpendicular to strike of the fault.***
- **Throw:** ***The vertical component measured in vertical plane containing the dip.***





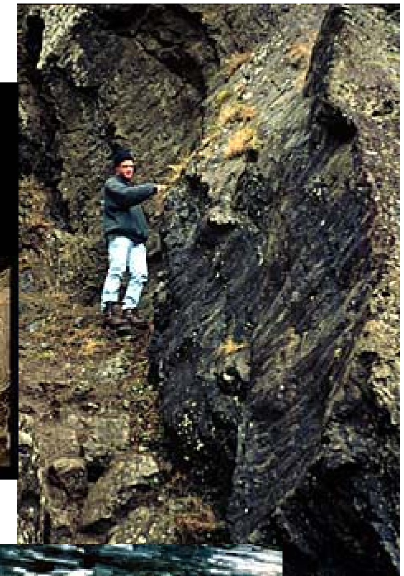
Faults Types

Features on the fault surface

- **Grooves (parallel to the movement direction)**
- **Growth of fibrous minerals (parallel to the movement direction)**
- **Slickensides are the polished fault surfaces.**
- **Small steps.**

All are considered a kind of lineation. They indicate the movement relative trend NW, NE … etc.

Small steps may also be used to determine the movement direction and direction of movement of the opposing wall. Slicklines usually record only the last moment event on the fault.

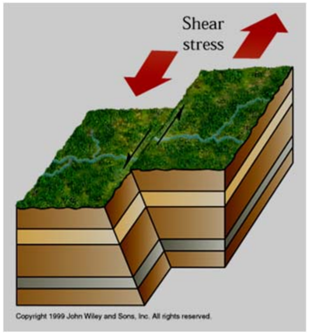


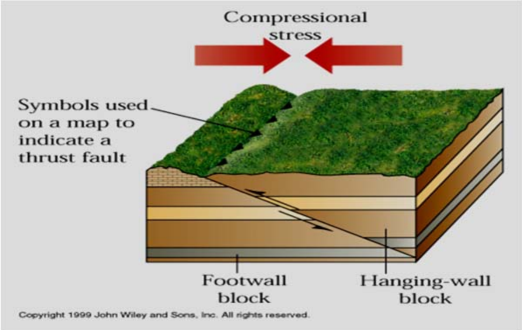
ANDERSON FAULTS CLASSIFICATION

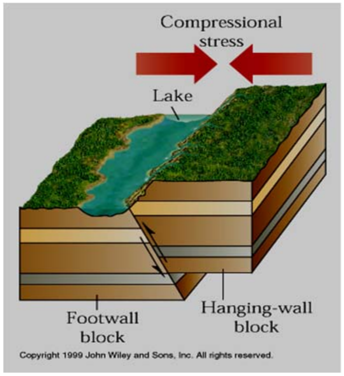
Anderson (1942) defined three types of faults:

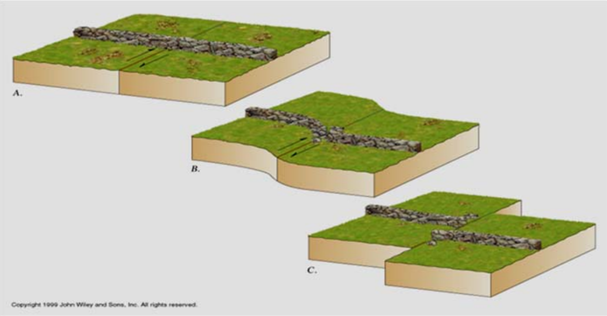
- ***Normal Faults***
- ***Thrust Faults***
- ***Wrench Faults (strike slip)***

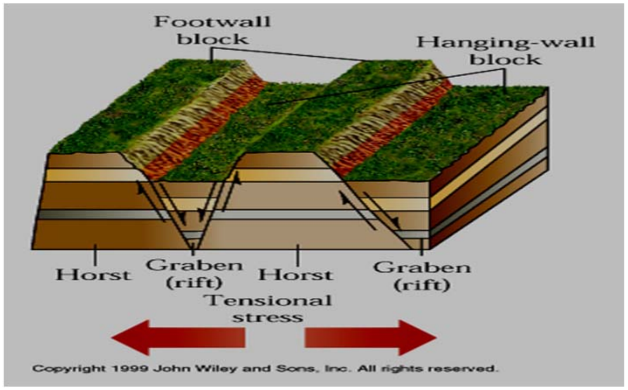
Different Type of Faults











Normal Fault

Normal Fault: **The hanging wall has moved down relative to the footwall.**

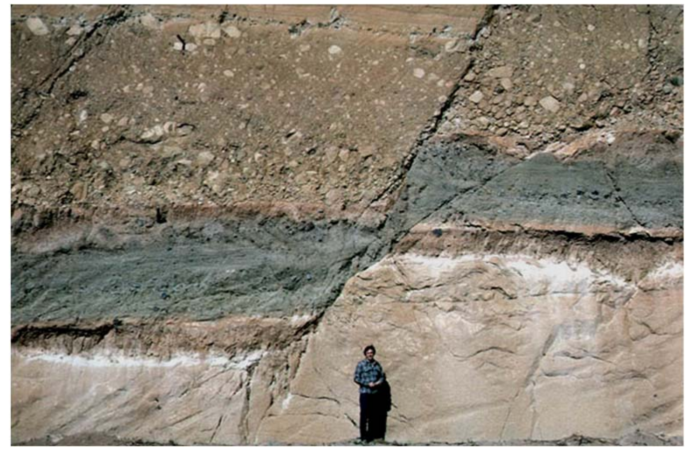
Graben: consists of a block that has dropped down between two subparllel normal faults that dip towards each other.

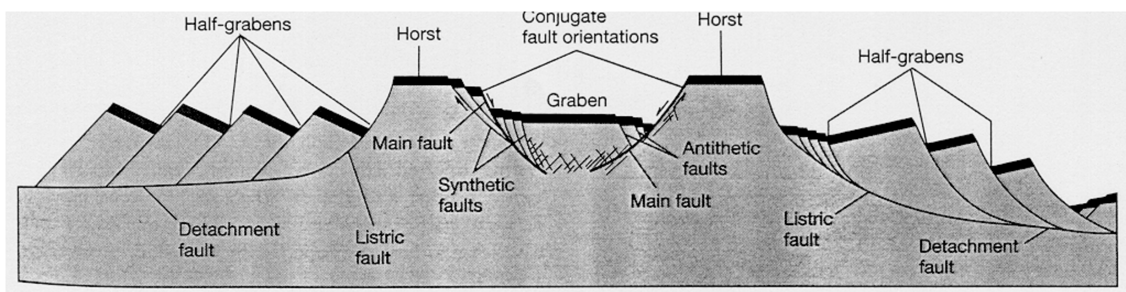
Horst : consists of two subparallel normal faults that dip away from each other so that the block between the two faults remains high.

Listric: are normal faults that frequently exhibit (concave-up) geometry so that they exhibit steep dip near surface and flatten with depth.

Normal faults usually found in areas where extensional regime is present.

Normal Faults





Thrust Fault

Thrust Faults: **In the thrust faults the hanging wall has moved up relative to the footwall (dip angle 30º or less)**

Reverse Faults: **Are similar to the thrust faults regarding the sense of motion but the dip angle of the fault plane is 30º or more**

Thrust faults usually formed in areas of comperssional regime.

Thrust Faults

Strike-Slip Fault

Strike-slip Faults: **Are faults that have movement along strikes.**

There are two types of strike slip faults:

A] Right lateral strike-slip fault (dextral): Where the side opposite the observer moves to the right.

B] Left lateral strike-slip fault (sinistral): Where the side opposite the observer moves to the left.

Note that the same sense of movement will also be observed from the other side of the fault.

Transform Faults

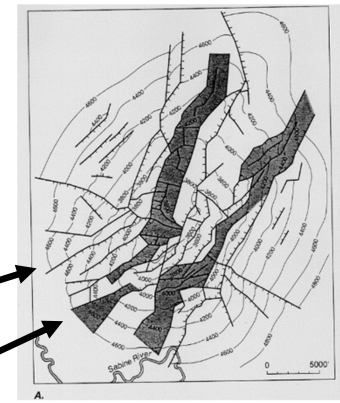
Transform Faults: Are a type of strike-slip fault (defined by Wilson 1965). They form due to **the differences in motion between lithospheric plates.** **They are basically occur where type of plate boundary is transformed into another.**

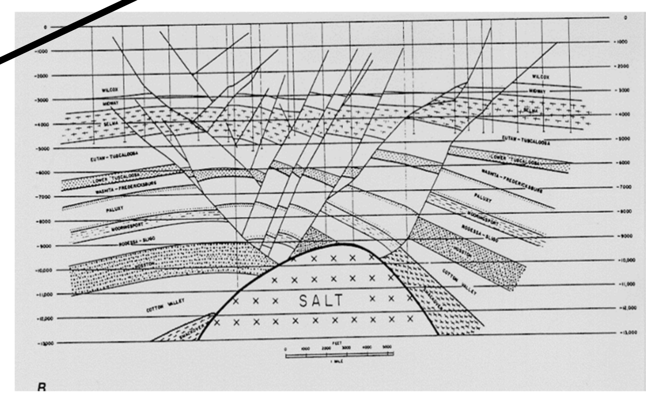
Main types of transform faults are:

- ***Ridge-Ridge***
- ***Ridge-Arc***
- ***Arc-Arc***

Other types of fault

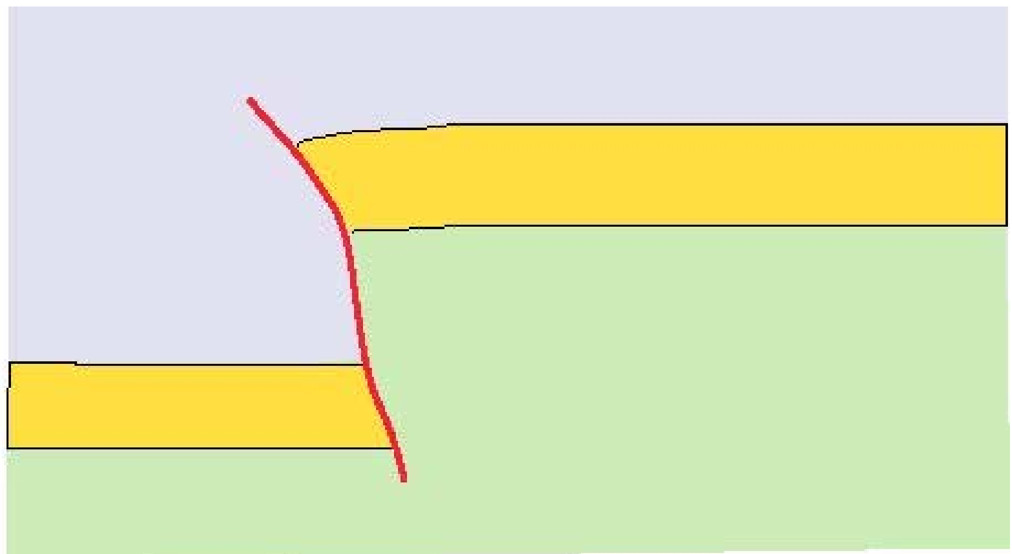
- ***en-echelon faults:*** Faults that are approximately parallel one another but occur in short unconnected segments, and sometimes overlapping.
- ***Radial faults:*** faults that are converge toward one point
- ***Concentric faults:*** faults that are concentric to a point.
- ***Bedding faults (bedding plane faults):*** follow bedding or occur parallel to the orientation of bedding planes.



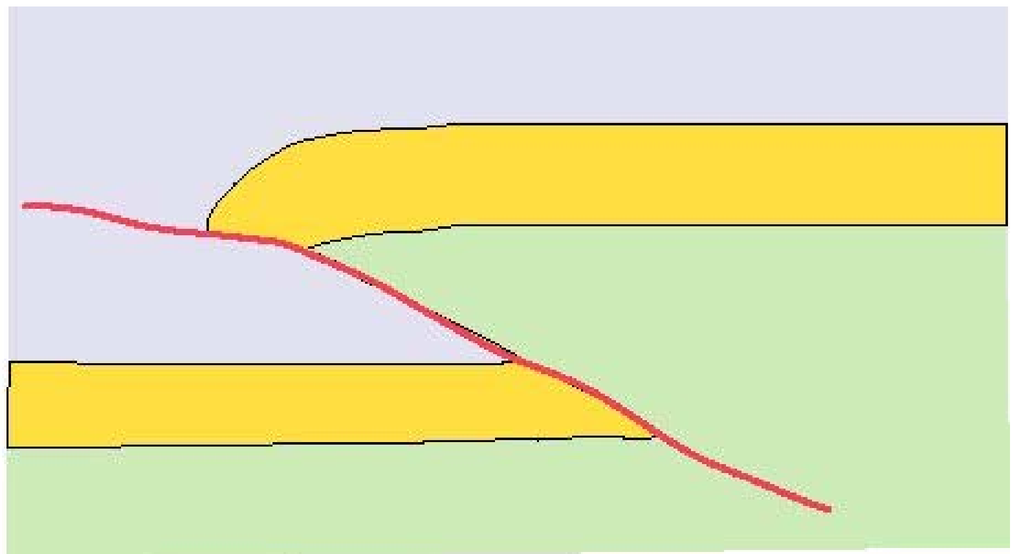


CRITERIA FOR FAULTING

- ***Repetition or omission** of stratigraphic units asymmetrical repetition*
- ***Displacement of recognizable marker** such as fossils, color, composition, texture ..etc.).*
- ***Truncation of structures**, beds or rock units.*
- ***Occurrence of fault rocks** (mylonite or cataclastic or both)*
- ***Presence of S or C structures** or both, rotated porphyry clasts and other evidence of shear zone.*
- ***Abundant veins**, silicification or other mineralization along fracture may indicate faulting.*
- ***Drag Units** appear to be pulled into a fault during movement (usually within the drag fold and the result is thrust fault)*
- ***Reverse drag** occurs along listric normal faults.*
- ***Slickensides** and slickenlines along a fault surface*
- ***Topographic characteristics** such as drainges that are controlled by faults and fault scarps.*



First-pass interp - steep faults..…not in vogue



Low-angle thrust (detachment at depth??)

Introduction
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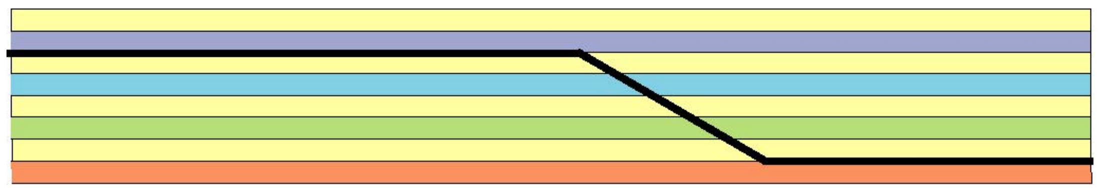
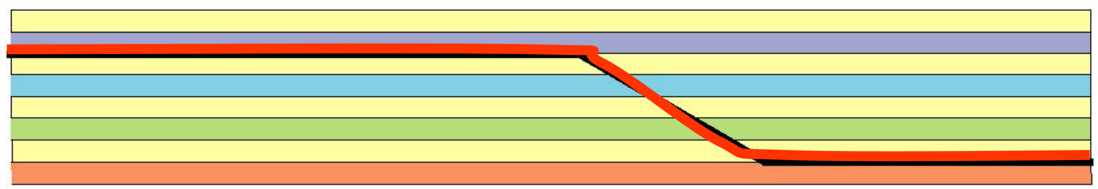
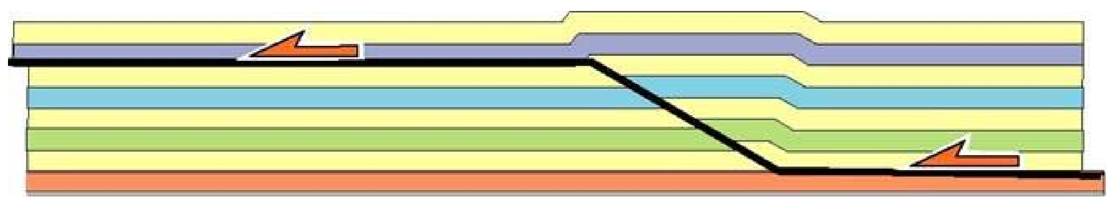
Trishear

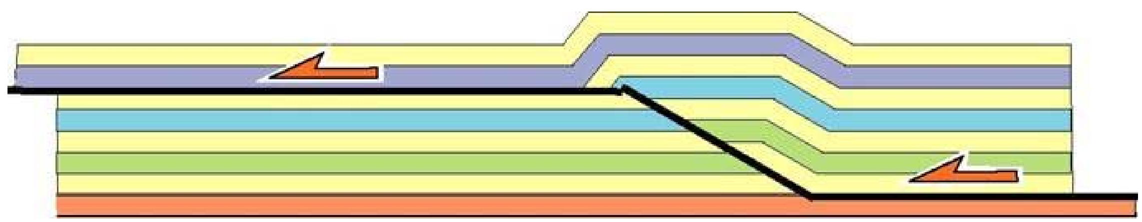
Simple fault-bend fold

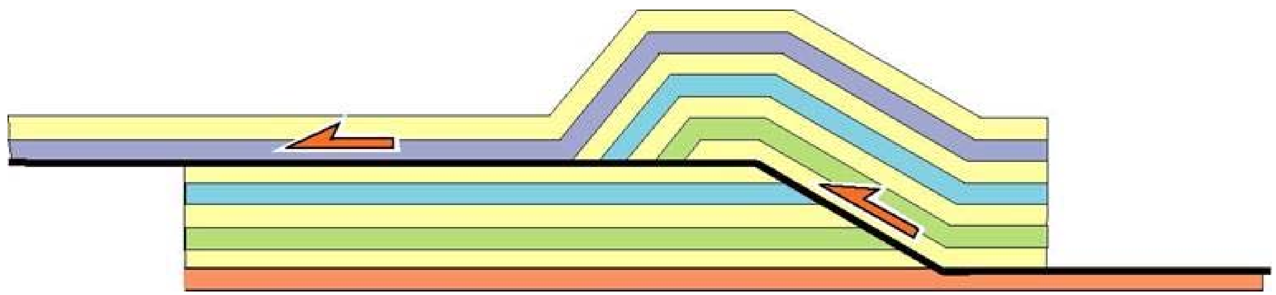
Suppe model





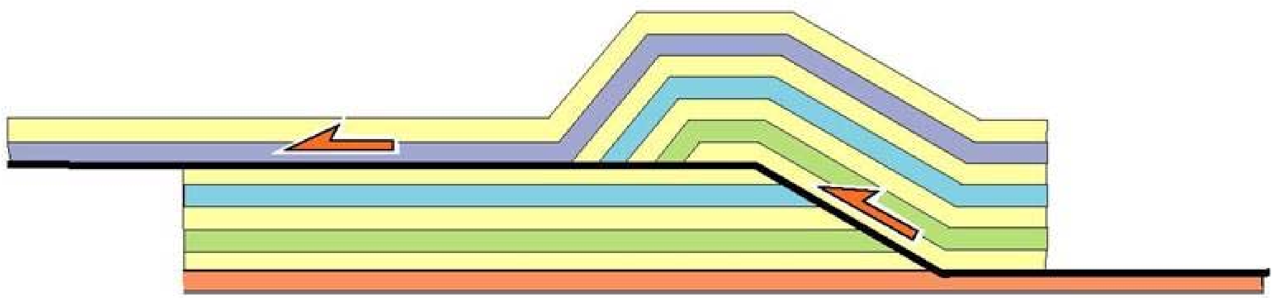






Thrust shape$_2$

Fold created in hanging-wall

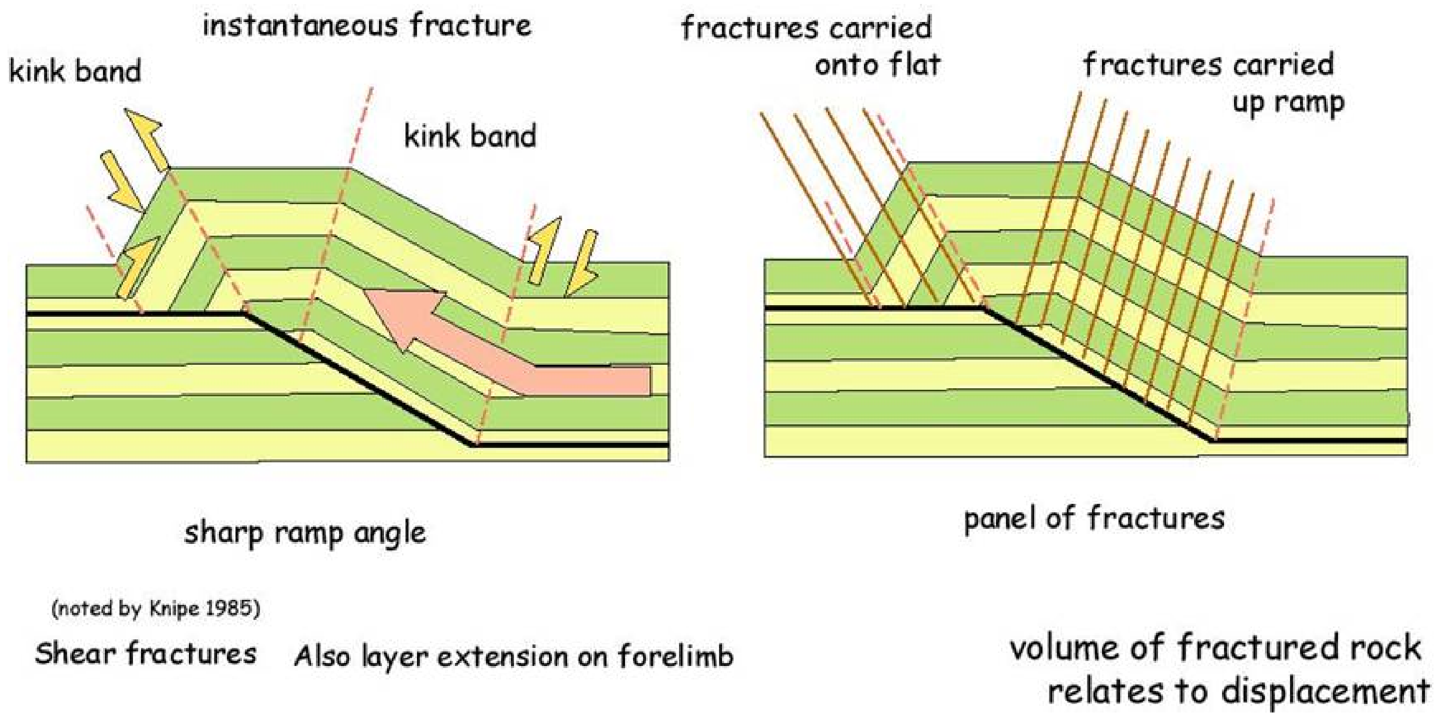


footwall remains undeformed

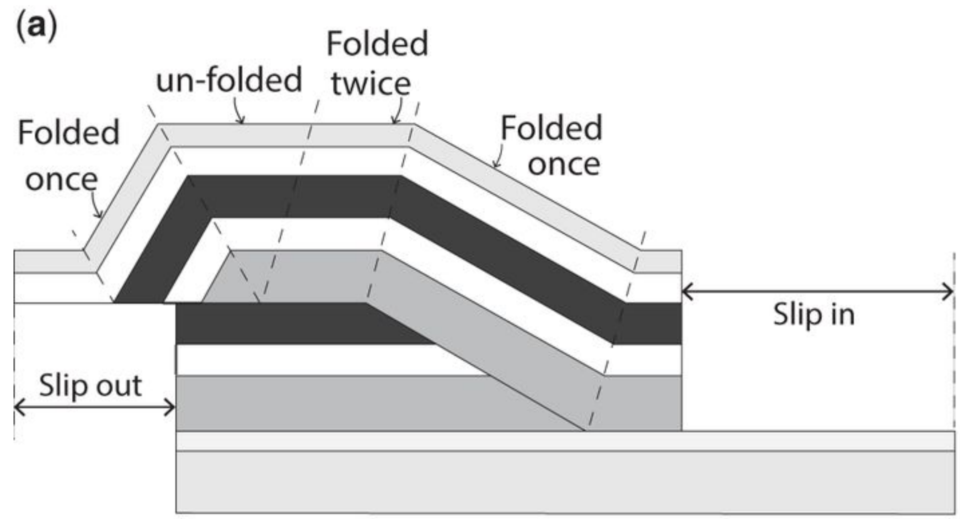
there are various relationships created between bedding on either side of the fault

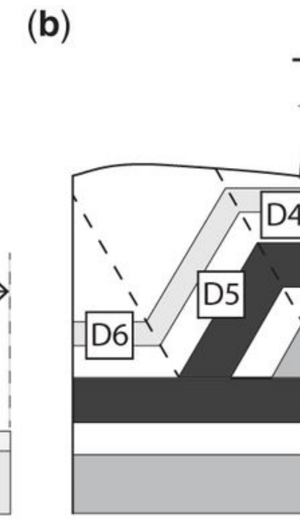
Image generated in Rampe 3.1 (Eric Mercier)

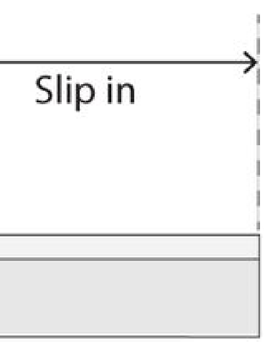
Suppe fault bend folding…..

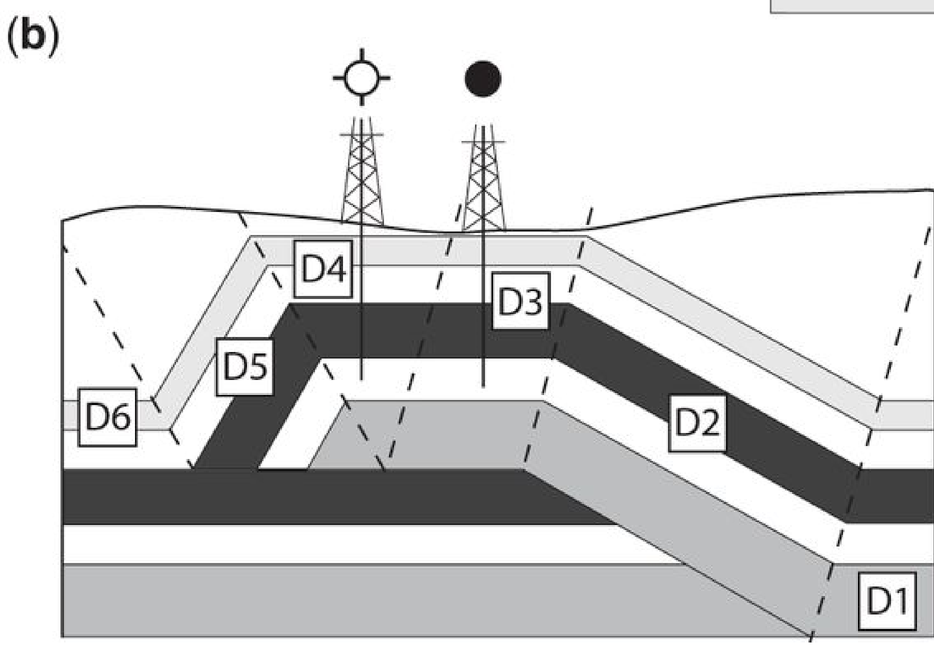


But is this too simplistic a style?


(a)
Folded
twice
un-folded
Folded
once
Folded
once
Slip in
Slip out


(b)
D4
D5
D6

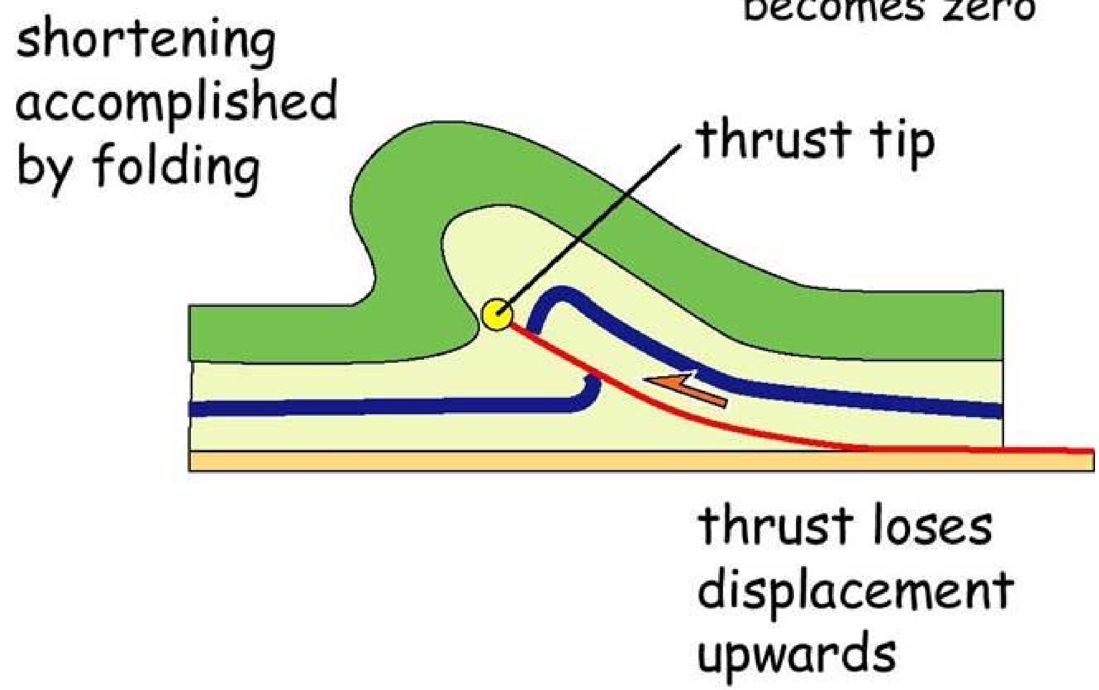

Slip in


(b)
D4
D3
D5
D6
D2
D1

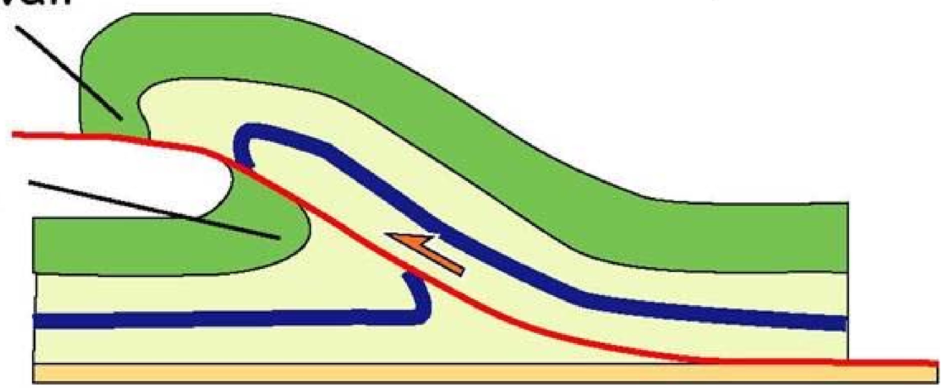

looking south
HW
ramp
footwall flat

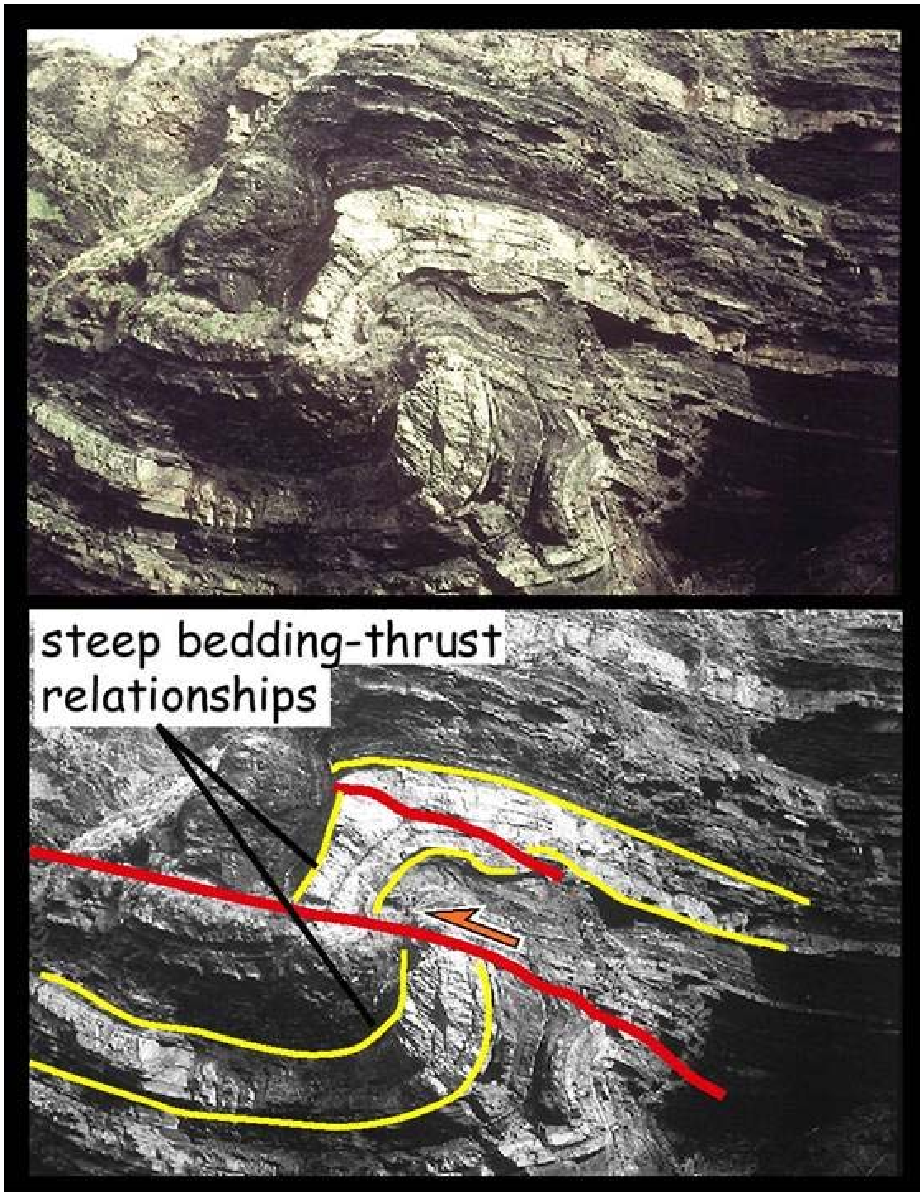
tip-line folds

tip = edge of fault plane,
where displacement
becomes zero

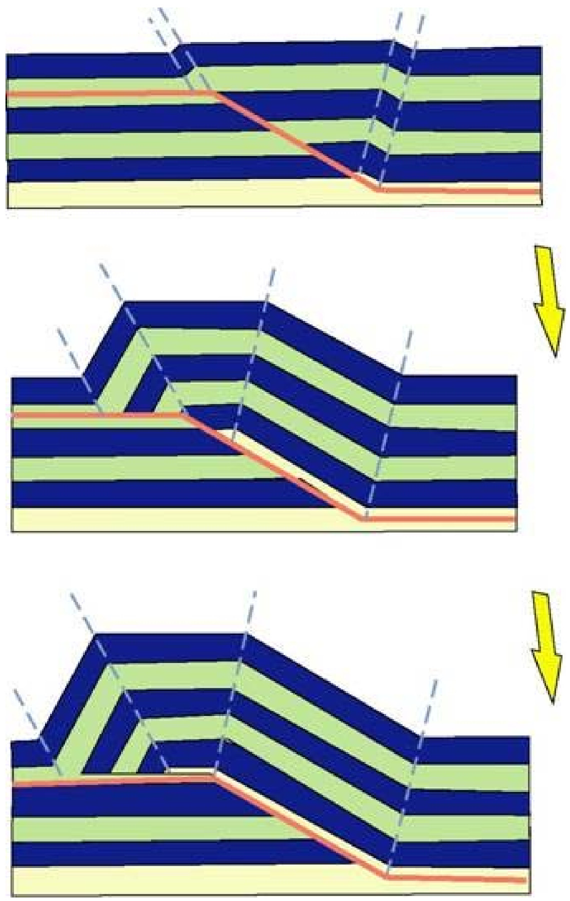


tip-line folds 2

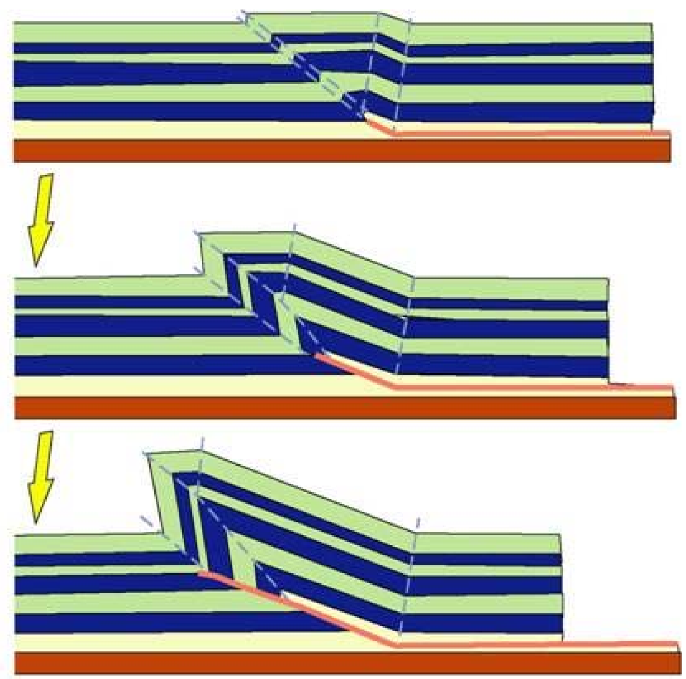



steep bedding-thrust
relationships

Panels of constant-dip strata, folds grow by hinge migration

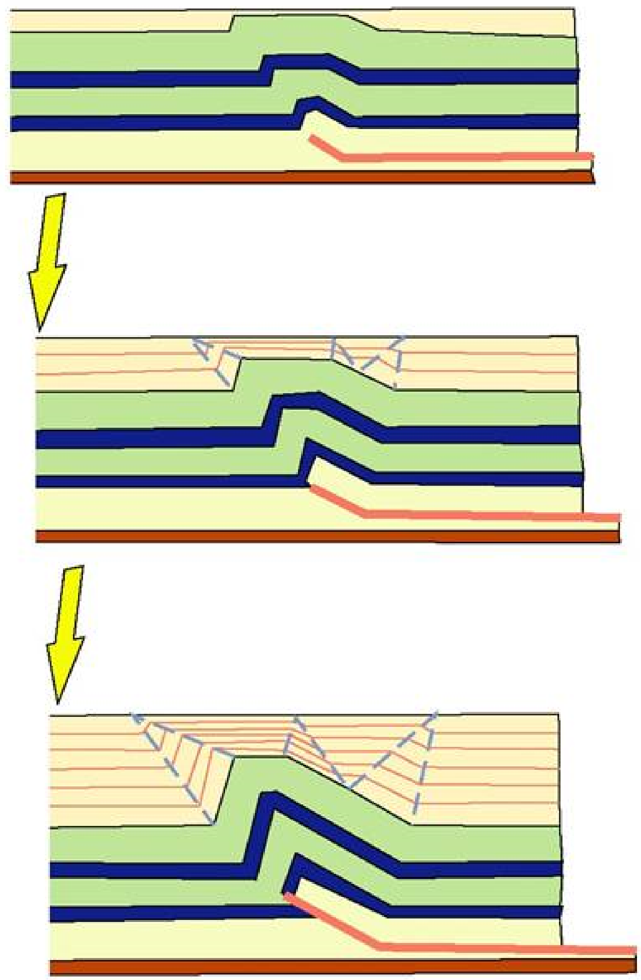


Fault-bend folding



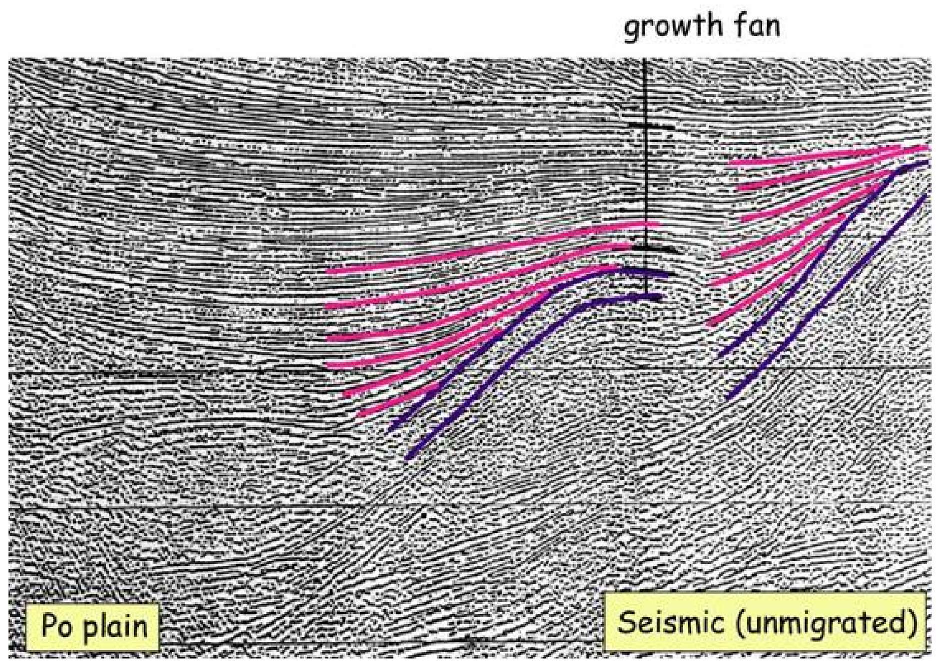
Fault-propagation folding

Growth patterns for fault propagation folds



Test kinematic models using growth strata

Growth triangles



“growth fans” rather than “growth triangles are the usual style….

So are the fault-bend models applicable?

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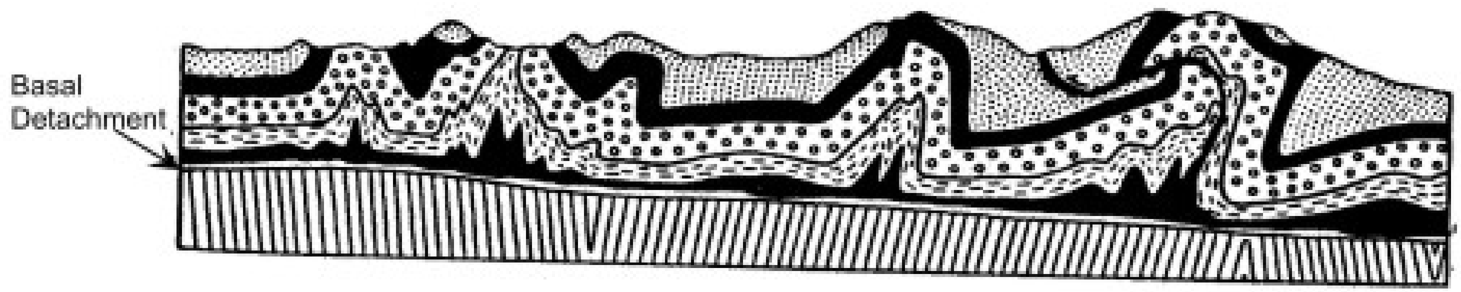
JURA MOUNTAINS



Detachment fold